

# William Kwako

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## EDUCATION

M.S., Computer Science – Interactive Intelligence, Georgia Tech, GPA: 4.0

August 2023 - August 2026

B.A., Physics, Willamette University

May 2019

## PROFESSIONAL EXPERIENCE

**Technical Solutions Engineer** – Epic Systems Corporation

December 2019 – December 2022

- Served as point of contact for all claims-related issues across five health system clients; led weekly and biweekly meetings with revenue cycle analysts and managers to triage and resolve issues
- Deployed, maintained, and debugged revenue cycle software, including HIPAA-mandated EDI claims and remittance transactions (837I/837P, 835)
- Coordinated large-scale claims implementations (claim attachments, PLBs, rapid reconciliation, state reporting) to meet standardized claim conformance requirements
- Wrote M/MUMPS queries against production healthcare databases containing PHI to diagnose client issues and analyze data for resolution
- Headed the Debugging Task Force; trained new hires on internal debugging tools and revised the onboarding curriculum

**Teaching Assistant** – Willamette University, Physics Department

January 2019 – May 2019

- Guided students through computational and experimental Introductory Physics 2 laboratory exercises
- Answered questions about class material and evaluated students' performance

**Intern** – Willamette University, Physics Department

May 2018 – July 2018

- Implemented 3D wavelet convolutions via FFT in Python to search for baryon acoustic oscillation (BAO) signals in the 2MASS Redshift Survey – the first application of this method to the 2MRS
- Applied Gaussian smoothing, zero-padding, and a custom local maxima detection algorithm to identify candidate BAO locations from a cosmological density field
- Built a computer for parallel GPU processing to support analysis of a large-scale cosmological dataset

## PROJECTS AND RESEARCH

- **Predicting Speedrunner Retention from Activity Patterns**

*Applied Machine Learning Research, 2026*

- Built an end-to-end pipeline predicting speedrunner retention, pulling data from the speedrun.com API into SQLite and engineering behavioral features (recency, frequency, consistency, run density)
- Compared three model families (logistic regression, gradient boosting, and a PyTorch MLP) across three games, reaching ~0.79 AUC on splits controlled for temporal and identity leakage
- **Tools:** Python, PyTorch, scikit-learn, NumPy, SQLite

- **Benchmarking Reinforcement and Imitation Learning on a Deterministic Platformer: Jump King**  
*Reinforcement Learning Research, 2026*
  - Built a full pipeline interfacing with a live commercial game through real-time state extraction (custom C# mod), TCP socket communication, and a custom Gymnasium environment
  - Designed and implemented behavioral cloning (BC) from human playthroughs, per-level reward shaping, and a class-based framework for training and evaluating RL, BC, and BC+RL agents
  - Chained per-level agents reliably complete the full 43-level base game in ~10 minutes
  - Showed BC initialization solves levels where pure RL fails entirely, and converges far faster elsewhere
  - **Tools:** Python, C#, Stable-Baselines3, Gymnasium, dnSpy, .NET
- **Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation**  
*Cosmology Research – Senior Thesis, Willamette University, 2018-2019*
  - Applied wavelet transformations to identify baryon acoustic oscillations (BAOs) in galaxy distribution data from the 2MASS Redshift Survey (2MRS)
  - Identified three BAO signals with only 0.0008% probability of occurring by chance ( $4.31\sigma$ )
  - **Tools:** Python, NumPy, SciPy, Matplotlib
- **Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems**  
*Educational Technology Research, 2025-2026*
  - Built a web application generating personalized coding exercises to improve code comprehension
  - Designed and implemented a multi-model pipeline with conditional routing, prompt chaining, guardrails, and feedback loops across OpenAI and Anthropic APIs
  - Built a validation and repair loop that executes generated code to confirm it runs and uses AST inspection to verify it meets user specifications, and escalates to a stronger model on failure
  - **Tools:** Python, Django, JavaScript, HTML/CSS, OpenAI and Anthropic APIs
- **Improving Users' Academic Writing Ability Through LLM-Generated Feedback**  
*Educational Technology Research, 2026*
  - Built a Django web application that generates writing feedback by decomposing analysis across six specialized LLM agents using LangGraph, with a synthesis step reconciling critiques
  - Designed a controlled evaluation (position-bias and cross-family-judge controls) that isolated analyzer strength as the performance bottleneck
  - Implemented RAG with Chroma (genre exemplars, rubric criteria) and tested it via ablation; found no net benefit at  $n=10$  and reported it as a null result
  - **Tools:** Python, Django, LangGraph, Chroma, HTML/CSS, OpenAI and Anthropic APIs

## POSTERS AND PRESENTATIONS

- *Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems*  
Poster Presentation – National Council on Measurement in Education (April 2026)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*  
Oral Presentation – Willamette-Linfield intercollegiate symposium (May 2019)  
Poster Presentations – 223<sup>rd</sup> Meeting of the American Astronomical Society (January 2019),  
Murdock College Science Research Conference (November 2018)

## TECHNICAL SKILLS

- **Languages:** Python, C#, HTML, CSS, JavaScript, SQL, M/MUMPS
- **AI & ML:** NumPy, Stable-Baselines3, Gymnasium, PyTorch, scikit-learn, OpenAI API, Anthropic API, Prompt Engineering, LangGraph, Chroma, RAG
- **Frameworks & Tools:** Django, Git, Unity, LaTeX, .NET, Android Studio, dnSpy